SQL INPUT

```
CREATE DATABASE IF NOT EXISTS RestaurantDB;
USE RestaurantDB;

CREATE TABLE Customer (
    customer_id INT PRIMARY KEY,
    name VARCHAR(50) NOT NULL,
    phone VARCHAR(15) UNIQUE,
    email VARCHAR(100)
) ENGINE=InnoDB;

CREATE TABLE Employee (
    employee_id INT PRIMARY KEY,
    name VARCHAR(50) NOT NULL,
    role VARCHAR(30) NOT NULL,
    phone VARCHAR(15) UNIQUE,
    salary DECIMAL(10,2) NOT NULL,
    CHECK (salary >= 0)
) ENGINE=InnoDB;

CREATE TABLE Menu (
    item_id INT PRIMARY KEY,
    item_name VARCHAR(50) NOT NULL,
    category VARCHAR(30) NOT NULL,
    price DECIMAL(8,2) NOT NULL,
    availability VARCHAR(10) DEFAULT 'Yes',
    CHECK (price >= 0),
    CHECK (availability IN ('Yes','No'))
) ENGINE=InnoDB;

CREATE TABLE Orders (
    order_id INT PRIMARY KEY,
    customer_id INT NOT NULL,
    employee_id INT,
    order_date DATETIME DEFAULT CURRENT_TIMESTAMP,
    status VARCHAR(20) DEFAULT 'Pending',
    FOREIGN KEY (customer_id) REFERENCES Customer(customer_id),
    FOREIGN KEY (employee_id) REFERENCES Employee(employee_id),
    CHECK (status IN ('Pending','Preparing','Completed','Cancelled'))
) ENGINE=InnoDB;

CREATE TABLE OrderDetails (
    order_detail_id INT PRIMARY KEY,
    order_id INT NOT NULL,
    item_id INT NOT NULL,
    quantity INT NOT NULL,
    item_price DECIMAL(8,2) NOT NULL,
    FOREIGN KEY (order_id) REFERENCES Orders(order_id),
    FOREIGN KEY (item_id) REFERENCES Menu(item_id),
    CHECK (quantity > 0),
    CHECK (item_price >= 0)
) ENGINE=InnoDB;

CREATE TABLE Payment (
    payment_id INT PRIMARY KEY,
    order_id INT NOT NULL,
    amount DECIMAL(10,2) NOT NULL,
    payment_method VARCHAR(20) NOT NULL,
    payment_status VARCHAR(20) DEFAULT 'Pending',
    payment_date DATETIME DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (order_id) REFERENCES Orders(order_id),
    CHECK (amount >= 0),
    CHECK (payment_method IN ('Cash','UPI','Card')),
    CHECK (payment_status IN ('Pending','Paid','Failed','Refunded'))
) ENGINE=InnoDB;

SHOW TABLES;
```

MYSQL OUTPUT

```
mysql> SHOW TABLES;
+---------------------------+
```

```
| Tables_in_restaurantdb    |
+---------------------------+
| Customer                  |
| Employee                  |
| Menu                      |
| Orders                    |
| OrderDetails              |
| Payment                   |
+---------------------------+
6 rows in set (0.00 sec)
```

SQL INPUT

```
INSERT INTO Customer VALUES
(1,'Aarav Sharma','9876543210','aarav@gmail.com'),
(2,'Priya Patil','9876543211','priya@gmail.com'),
(3,'Rahul Verma','9876543212','rahul@gmail.com'),
(4,'Sneha Joshi','9876543213','sneha@gmail.com'),
(5,'Rohan Mehta','9876543214','rohan@gmail.com');

INSERT INTO Employee VALUES
(201,'Amit Kulkarni','Manager','9123456780',35000.00),
(202,'Neha Shah','Chef','9123456781',25000.00),
(203,'Karan Singh','Waiter','9123456782',15000.00),
(204,'Pooja Desai','Cashier','9123456783',18000.00),
(205,'Vikas Jain','Chef','9123456784',24000.00);

INSERT INTO Menu VALUES
(101,'Paneer Pizza','Pizza',250.00,'Yes'),
(102,'Veg Burger','Burger',120.00,'Yes'),
(103,'White Sauce Pasta','Pasta',180.00,'Yes'),
(104,'French Fries','Starter',90.00,'Yes'),
(105,'Cold Coffee','Beverage',100.00,'Yes'),
(106,'Chocolate Brownie','Dessert',130.00,'No');

INSERT INTO Orders
(order_id,customer_id,employee_id,order_date,status) VALUES
(1001,1,203,'2026-10-10 12:30:00','Completed'),
(1002,2,203,'2026-10-10 13:00:00','Completed'),
(1003,3,202,'2026-10-11 14:15:00','Preparing'),
(1004,1,203,'2026-10-11 19:00:00','Pending'),
(1005,4,204,'2026-10-12 12:00:00','Completed');

INSERT INTO OrderDetails VALUES
(1,1001,101,2,250.00),
(2,1001,104,1,90.00),
(3,1002,102,2,120.00),
(4,1002,105,2,100.00),
(5,1003,103,1,180.00),
(6,1003,104,2,90.00),
(7,1004,101,1,250.00),
(8,1005,102,1,120.00),
(9,1005,105,1,100.00);

INSERT INTO Payment
(payment_id,order_id,amount,payment_method,payment_status,payment_date) VALUES
(501,1001,590.00,'UPI','Paid','2026-10-10 12:45:00'),
(502,1002,440.00,'Cash','Paid','2026-10-10 13:15:00'),
(503,1005,220.00,'Card','Paid','2026-10-12 12:20:00');
```

MYSQL OUTPUT

```
Query OK, 5 rows affected (0.01 sec)
Query OK, 5 rows affected (0.01 sec)
Query OK, 6 rows affected (0.01 sec)
Query OK, 5 rows affected (0.01 sec)
Query OK, 9 rows affected (0.01 sec)
Query OK, 3 rows affected (0.01 sec)
```

SQL INPUT

```
SELECT * FROM Customer;
```

MYSQL OUTPUT

```
mysql> SELECT * FROM Customer;
+-------------+--------------+------------+-----------------+
| customer_id | name         | phone      | email           |
+-------------+--------------+------------+-----------------+
| 1           | Aarav Sharma | 9876543210 | aarav@gmail.com |
| 2           | Priya Patil | 9876543211 | priya@gmail.com |
| 3           | Rahul Verma | 9876543212 | rahul@gmail.com |
| 4           | Sneha Joshi | 9876543213 | sneha@gmail.com |
| 5           | Rohan Mehta | 9876543214 | rohan@gmail.com |
+-------------+--------------+------------+-----------------+
5 rows in set (0.00 sec)
```

SQL INPUT

```
SELECT * FROM Employee;
```

MYSQL OUTPUT

```
mysql> SELECT * FROM Employee;
+-------------+---------------+----------+------------+----------+
| employee_id | name          | role     | phone      | salary   |
+-------------+---------------+----------+------------+----------+
| 201         | Amit Kulkarni | Manager | 9123456780 | 35000.00 |
| 202         | Neha Shah     | Chef     | 9123456781 | 25000.00 |
| 203         | Karan Singh   | Waiter   | 9123456782 | 15000.00 |
| 204         | Pooja Desai   | Cashier | 9123456783 | 18000.00 |
| 205         | Vikas Jain    | Chef     | 9123456784 | 24000.00 |
+-------------+---------------+----------+------------+----------+
5 rows in set (0.00 sec)
```

SQL INPUT

```
SELECT * FROM Menu ORDER BY item_id;
```

MYSQL OUTPUT

```
mysql> SELECT * FROM Menu ORDER BY item_id;
+---------+-------------------+----------+--------+--------------+
| item_id | item_name         | category | price | availability |
+---------+-------------------+----------+--------+--------------+
| 101     | Paneer Pizza      | Pizza    | 250.00 | Yes          |
| 102     | Veg Burger        | Burger   | 120.00 | Yes          |
| 103     | White Sauce Pasta | Pasta    | 180.00 | Yes          |
| 104     | French Fries      | Starter | 90.00 | Yes           |
| 105     | Cold Coffee       | Beverage | 100.00 | Yes          |
| 106     | Chocolate Brownie | Dessert | 130.00 | No           |
+---------+-------------------+----------+--------+--------------+
6 rows in set (0.00 sec)
```

SQL INPUT

```
SELECT * FROM Orders ORDER BY order_id;
```

MYSQL OUTPUT

```
mysql> SELECT * FROM Orders ORDER BY order_id;
+----------+-------------+-------------+---------------------+-----------+
| order_id | customer_id | employee_id | order_date          | status    |
+----------+-------------+-------------+---------------------+-----------+
| 1001     | 1           | 203         | 2026-10-10 12:30:00 | Completed |
| 1002     | 2           | 203         | 2026-10-10 13:00:00 | Completed |
| 1003     | 3           | 202         | 2026-10-11 14:15:00 | Preparing |
| 1004     | 1           | 203         | 2026-10-11 19:00:00 | Pending   |
| 1005     | 4           | 204         | 2026-10-12 12:00:00 | Completed |
+----------+-------------+-------------+---------------------+-----------+
5 rows in set (0.00 sec)
```

SQL INPUT

```
SELECT * FROM OrderDetails ORDER BY order_detail_id;
```

MYSQL OUTPUT

```
mysql> SELECT * FROM OrderDetails ORDER BY order_detail_id;
+-----------------+----------+---------+----------+------------+
| order_detail_id | order_id | item_id | quantity | item_price |
+-----------------+----------+---------+----------+------------+
| 1               | 1001     | 101     | 2        | 250.00     |
| 2               | 1001     | 104     | 1        | 90.00      |
| 3               | 1002     | 102     | 2        | 120.00     |
| 4               | 1002     | 105     | 2        | 100.00     |
| 5               | 1003     | 103     | 1        | 180.00     |
| 6               | 1003     | 104     | 2        | 90.00      |
| 7               | 1004     | 101     | 1        | 250.00     |
| 8               | 1005     | 102     | 1        | 120.00     |
| 9               | 1005     | 105     | 1        | 100.00     |
+-----------------+----------+---------+----------+------------+
9 rows in set (0.00 sec)
```

SQL INPUT

```
SELECT * FROM Payment ORDER BY payment_id;
```

MYSQL OUTPUT

```
mysql> SELECT * FROM Payment ORDER BY payment_id;
+------------+----------+--------+----------------+----------------+---------------------+
| payment_id | order_id | amount | payment_method | payment_status | payment_date        |
+------------+----------+--------+----------------+----------------+---------------------+
| 501        | 1001     | 590.00 | UPI            | Paid           | 2026-10-10 12:45:00 |
| 502        | 1002     | 440.00 | Cash           | Paid           | 2026-10-10 13:15:00 |
| 503        | 1005     | 220.00 | Card           | Paid           | 2026-10-12 12:20:00 |
+------------+----------+--------+----------------+----------------+---------------------+
3 rows in set (0.00 sec)
```

SQL INPUT

```
CREATE TABLE TestTable (
    test_id INT PRIMARY KEY,
    test_name VARCHAR(50)
);

ALTER TABLE Customer ADD email2 VARCHAR(100);
ALTER TABLE Customer DROP COLUMN email2;
DESCRIBE Customer;

CREATE TABLE TestTruncate (
    test_id INT PRIMARY KEY,
    test_name VARCHAR(30)
);
INSERT INTO TestTruncate VALUES (1,'Test Record');
TRUNCATE TABLE TestTruncate;
DROP TABLE TestTruncate;
```

MYSQL OUTPUT

```
mysql> CREATE TABLE TestTable (...);
Query OK, 0 rows affected (0.02 sec)

mysql> ALTER TABLE Customer ADD email2 VARCHAR(100);
Query OK, 0 rows affected (0.02 sec)

mysql> ALTER TABLE Customer DROP COLUMN email2;
Query OK, 0 rows affected (0.02 sec)

mysql> TRUNCATE TABLE TestTruncate;
Query OK, 0 rows affected (0.01 sec)

mysql> DROP TABLE TestTruncate;
Query OK, 0 rows affected (0.01 sec)
```

SQL INPUT

```
INSERT INTO Customer
(customer_id,name,phone,email)
VALUES (6,'Ananya Deshmukh','9876543215','ananya@gmail.com');

UPDATE Menu SET price=270.00 WHERE item_id=101;
SELECT item_id,item_name,price FROM Menu WHERE item_id=101;

DELETE FROM Customer WHERE customer_id=6;
SELECT customer_id,name FROM Customer WHERE customer_id=6;
```

MYSQL OUTPUT

```
mysql> INSERT INTO Customer VALUES (6,'Ananya Deshmukh','9876543215','ananya@gmail.com');
Query OK, 1 row affected (0.01 sec)

mysql> UPDATE Menu SET price=270.00 WHERE item_id=101;
Query OK, 1 row affected (0.01 sec)

mysql> SELECT item_id,item_name,price FROM Menu WHERE item_id=101;
+---------+--------------+--------+
| item_id | item_name    | price |
+---------+--------------+--------+
| 101     | Paneer Pizza | 270.00 |
+---------+--------------+--------+
1 row in set (0.00 sec)

mysql> DELETE FROM Customer WHERE customer_id=6;
Query OK, 1 row affected (0.01 sec)

mysql> SELECT customer_id,name FROM Customer WHERE customer_id=6;
Empty set (0.00 sec)
```

SQL INPUT

```
START TRANSACTION;
INSERT INTO Customer(customer_id,name,phone)
VALUES (10,'Temporary Demo','9000000010');
COMMIT;

START TRANSACTION;
UPDATE Menu SET price=999.00 WHERE item_id=101;
ROLLBACK;
SELECT item_id,item_name,price FROM Menu WHERE item_id=101;

START TRANSACTION;
UPDATE Menu SET price=260.00 WHERE item_id=101;
SAVEPOINT S1;
UPDATE Menu SET price=150.00 WHERE item_id=102;
ROLLBACK TO SAVEPOINT S1;
COMMIT;
SELECT item_id,item_name,price FROM Menu WHERE item_id IN (101,102);
```

MYSQL OUTPUT

```
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

mysql> ROLLBACK;
Query OK, 0 rows affected (0.00 sec)

+---------+--------------+--------+
| item_id | item_name    | price |
+---------+--------------+--------+
| 101     | Paneer Pizza | 250.00 |
+---------+--------------+--------+
1 row in set (0.00 sec)

mysql> ROLLBACK TO SAVEPOINT S1;
Query OK, 0 rows affected (0.00 sec)

+---------+--------------+--------+
| item_id | item_name    | price |
+---------+--------------+--------+
| 101     | Paneer Pizza | 260.00 |
| 102     | Veg Burger   | 120.00 |
+---------+--------------+--------+
2 rows in set (0.00 sec)
```

SQL INPUT

```
SELECT item_name,price FROM Menu WHERE availability='Yes' ORDER BY price ASC;
SELECT item_name,price FROM Menu WHERE price BETWEEN 100 AND 250;
SELECT item_name FROM Menu WHERE item_name LIKE '%a%';
```

MYSQL OUTPUT

```
mysql> SELECT item_name,price FROM Menu WHERE availability='Yes' ORDER BY price ASC;
+------------------+--------+
| item_name        | price |
+------------------+--------+
| French Fries     | 90.00 |
| Cold Coffee      | 100.00 |
| Veg Burger       | 120.00 |
| White Sauce Pasta | 180.00 |
| Paneer Pizza     | 250.00 |
+------------------+--------+
5 rows in set (0.00 sec)

mysql> SELECT item_name,price FROM Menu WHERE price BETWEEN 100 AND 250;
+------------------+--------+
| item_name        | price |
+------------------+--------+
| Paneer Pizza     | 250.00 |
| Veg Burger       | 120.00 |
| White Sauce Pasta | 180.00 |
| Cold Coffee      | 100.00 |
| Chocolate Brownie | 130.00 |
+------------------+--------+
5 rows in set (0.00 sec)
```

SQL INPUT

```
-- INNER JOIN
SELECT c.name AS customer_name,o.order_id,o.status
FROM Customer c INNER JOIN Orders o
ON c.customer_id=o.customer_id ORDER BY o.order_id;

-- LEFT JOIN
SELECT c.name AS customer_name,o.order_id
FROM Customer c LEFT JOIN Orders o
ON c.customer_id=o.customer_id
ORDER BY c.customer_id,o.order_id;

-- RIGHT JOIN
SELECT m.item_name,od.order_id,od.quantity
FROM OrderDetails od RIGHT JOIN Menu m
ON od.item_id=m.item_id
ORDER BY m.item_id,od.order_id;
```

MYSQL OUTPUT

```
mysql> INNER JOIN result
+---------------+----------+-----------+
| customer_name | order_id | status    |
+---------------+----------+-----------+
| Aarav Sharma | 1001      | Completed |
| Priya Patil   | 1002     | Completed |
| Rahul Verma   | 1003     | Preparing |
| Aarav Sharma | 1004      | Pending   |
| Sneha Joshi   | 1005     | Completed |
+---------------+----------+-----------+
5 rows in set (0.00 sec)

mysql> LEFT JOIN result
+---------------+----------+
| customer_name | order_id |
+---------------+----------+
| Aarav Sharma | 1001      |
| Aarav Sharma | 1004      |
| Priya Patil   | 1002     |
| Rahul Verma   | 1003     |
| Sneha Joshi   | 1005     |
| Rohan Mehta   | NULL     |
+---------------+----------+
6 rows in set (0.00 sec)

mysql> RIGHT JOIN result
+-------------------+----------+----------+
| item_name         | order_id | quantity |
+-------------------+----------+----------+
| Paneer Pizza      | 1001     | 2        |
| Paneer Pizza      | 1004     | 1        |
| Veg Burger        | 1002     | 2        |
| Veg Burger        | 1005     | 1        |
| White Sauce Pasta | 1003     | 1        |
| French Fries      | 1001     | 1        |
| French Fries      | 1003     | 2        |
| Cold Coffee       | 1002     | 2        |
| Cold Coffee       | 1005     | 1        |
| Chocolate Brownie | NULL     | NULL     |
+-------------------+----------+----------+
10 rows in set (0.00 sec)
```

SQL INPUT

```
SELECT c.name AS customer_name,o.order_id,m.item_name,
       od.quantity,od.item_price,
       od.quantity*od.item_price AS line_total
FROM Customer c
JOIN Orders o ON c.customer_id=o.customer_id
JOIN OrderDetails od ON o.order_id=od.order_id
JOIN Menu m ON od.item_id=m.item_id
WHERE o.order_id=1001;
```

MYSQL OUTPUT

```
mysql> SELECT ... WHERE o.order_id=1001;
+---------------+----------+--------------+----------+------------+------------+
| customer_name | order_id | item_name    | quantity | item_price | line_total |
+---------------+----------+--------------+----------+------------+------------+
| Aarav Sharma | 1001      | Paneer Pizza | 2        | 250.00     | 500.00     |
| Aarav Sharma | 1001      | French Fries | 1        | 90.00      | 90.00      |
+---------------+----------+--------------+----------+------------+------------+
2 rows in set (0.00 sec)
```

SQL INPUT

```
DELIMITER //
CREATE PROCEDURE ShowOrderBill(IN p_order_id INT)
BEGIN
    SELECT o.order_id,c.name AS customer_name,
            COALESCE(SUM(od.quantity*od.item_price),0) AS total_bill
    FROM Orders o
    JOIN Customer c ON o.customer_id=c.customer_id
    LEFT JOIN OrderDetails od ON o.order_id=od.order_id
    WHERE o.order_id=p_order_id
    GROUP BY o.order_id,c.name;
END //
DELIMITER ;
CALL ShowOrderBill(1001);
```

MYSQL OUTPUT

```
mysql> CALL ShowOrderBill(1001);
+----------+---------------+------------+
| order_id | customer_name | total_bill |
+----------+---------------+------------+
| 1001     | Aarav Sharma | 590.00      |
+----------+---------------+------------+
1 row in set (0.00 sec)
```

SQL INPUT

```
DELIMITER //
CREATE FUNCTION CalculateBill(p_order_id INT)
RETURNS DECIMAL(10,2)
READS SQL DATA
BEGIN
    DECLARE total DECIMAL(10,2);
    SELECT COALESCE(SUM(quantity*item_price),0)
    INTO total FROM OrderDetails
    WHERE order_id=p_order_id;
    RETURN total;
END //
DELIMITER ;

SELECT CalculateBill(1001) AS total_bill;
SELECT order_id,CalculateBill(order_id) AS total_bill
FROM Orders ORDER BY order_id;
```

MYSQL OUTPUT

```
mysql> SELECT CalculateBill(1001) AS total_bill;
+------------+
| total_bill |
+------------+
| 590.00     |
+------------+
1 row in set (0.00 sec)

mysql> SELECT order_id,CalculateBill(order_id) AS total_bill FROM Orders;
+----------+------------+
| order_id | total_bill |
+----------+------------+
| 1001     | 590.00     |
| 1002     | 440.00     |
| 1003     | 360.00     |
| 1004     | 250.00     |
| 1005     | 220.00     |
+----------+------------+
5 rows in set (0.00 sec)
```

SQL INPUT

```
DELIMITER //
CREATE TRIGGER CheckMenuAvailability
BEFORE INSERT ON OrderDetails
FOR EACH ROW
BEGIN
    DECLARE item_status VARCHAR(10);
    SELECT availability INTO item_status
    FROM Menu WHERE item_id=NEW.item_id;
    IF item_status IS NULL THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT='Menu item does not exist';
    ELSEIF item_status <> 'Yes' THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT='Food item is not available';
    END IF;
END //
DELIMITER ;

INSERT INTO OrderDetails VALUES (10,1004,106,1,130.00);
```

MYSQL OUTPUT

```
mysql> INSERT INTO OrderDetails VALUES (10,1004,106,1,130.00);
ERROR 1644 (45000): Food item is not available
```

SQL INPUT

```
DELIMITER //
CREATE PROCEDURE DisplayEmployeeNames()
BEGIN
    DECLARE finished INT DEFAULT 0;
    DECLARE emp_name VARCHAR(50);
    DECLARE all_names TEXT DEFAULT '';
    DECLARE emp_cursor CURSOR FOR
        SELECT name FROM Employee ORDER BY employee_id;
    DECLARE CONTINUE HANDLER FOR NOT FOUND SET finished=1;
    OPEN emp_cursor;
    read_loop: LOOP
        FETCH emp_cursor INTO emp_name;
        IF finished=1 THEN LEAVE read_loop; END IF;
        SET all_names=CONCAT(all_names,
             IF(all_names='','',', '),emp_name);
    END LOOP;
    CLOSE emp_cursor;
    SELECT all_names AS employee_names;
END //
DELIMITER ;
CALL DisplayEmployeeNames();
```

MYSQL OUTPUT

```
mysql> CALL DisplayEmployeeNames();
+-------------------------------------------------------------+
| employee_names                                              |
+-------------------------------------------------------------+
| Amit Kulkarni, Neha Shah, Karan Singh, Pooja Desai, Vikas Jain |
+-------------------------------------------------------------+
1 row in set (0.00 sec)
```

SQL INPUT

```
SELECT SUM(amount) AS total_revenue
FROM Payment WHERE payment_status='Paid';

SELECT status,COUNT(*) AS total_orders
FROM Orders GROUP BY status ORDER BY status;

SELECT item_name,price FROM Menu
WHERE price>(SELECT AVG(price) FROM Menu)
ORDER BY price DESC;
```

MYSQL OUTPUT

```
mysql> SELECT SUM(amount) AS total_revenue FROM Payment WHERE payment_status='Paid';
+---------------+
| total_revenue |
+---------------+
| 1250.00       |
+---------------+
1 row in set (0.00 sec)

mysql> SELECT status,COUNT(*) AS total_orders FROM Orders GROUP BY status;
+-----------+--------------+
| status    | total_orders |
+-----------+--------------+
| Completed | 3            |
| Pending   | 1            |
| Preparing | 1            |
+-----------+--------------+
3 rows in set (0.00 sec)

mysql> SELECT item_name,price FROM Menu WHERE price>(SELECT AVG(price) FROM Menu);
+------------------+--------+
| item_name        | price |
+------------------+--------+
| Paneer Pizza     | 250.00 |
| White Sauce Pasta | 180.00 |
+------------------+--------+
2 rows in set (0.00 sec)
```